airbnb

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Grupprojekt

DATASET: INSIDE AIRBNB, STOCKHOLM LISTINGS
4 955 LISTNINGAR, 79 KOLUMNER

MÅL: PREDICERA PRIS PER NATT OCH ANALYSERA VAD SOM DRIVER PRISET

Datarening och EDA

- Ladda data, inspektera, kör lite statistik för att börja bekanta oss med datasetet.

```python
import pandas as pd

# Bekanta oss med datasetet!

df = pd.read_csv('data/airbnb_listings_stockholm_detailed.csv')

print("df.shape")
print(df.shape)

print("5 första posterna (df.head()):")
print(df.head())

print("Statistik på alla numeriska kolumneriska kolumner (df.describe):")
print(df.describe())          # statistik på alla numeriska kolumner

print("Hur många observationer per område (df['neighbourhood'])")
print(df['neighbourhood'].value_counts())  # hur många observationer i varje område

print("df.info()")
print(df.info())
```

```
df.shape
(4955, 79)
5 första posterna (df.head()):
        id                            listing_url       scrape_id last_scraped  \
0  164448  https://www.airbnb.com/rooms/164448  20250929042406   2025-09-29
1  220851  https://www.airbnb.com/rooms/220851  20250929042406   2025-09-29
2  230411  https://www.airbnb.com/rooms/230411  20250929042406   2025-09-29
3  242188  https://www.airbnb.com/rooms/242188  20250929042406   2025-09-29
4  273906  https://www.airbnb.com/rooms/273906  20250929042406   2025-09-29

       source                                       name  \
0  city scrape  Double room in central Stockholm with Wi-Fi
1  city scrape                          One room in appartement
2  city scrape           Cozy apartment in central Stockholm
3  city scrape  Single room in central Stockholm with Wi-Fi
4  city scrape              Penthouse in central Stockholm

                                         description  \
0  I am renting out a nice double room on the top...
1                                            welcome!
2                                                 NaN
3  I am renting out a nice single room on the top...
4                                                 NaN

                                neighborhood_overview  \
0                                                 NaN
1  Many restaurangs wery close and walkingdistanc...
2  Restaurants, cafés, museums, art galleries, pa...
3                                                 NaN
4                                                 NaN

                                           picture_url  host_id  ...  \
0  https://a0.muscache.com/pictures/f56d8d10-a7fa...    784312  ...
1  https://a0.muscache.com/pictures/2085606/7a706...    412283  ...
2  https://a0.muscache.com/pictures/2806060/7fc68...   1250232  ...
3  https://a0.muscache.com/pictures/2303148/55d1e...    784312  ...
4  https://a0.muscache.com/pictures/2881091/f5404...   1432722  ...

  review_scores_communication review_scores_location review_scores_value  \
0                        4.97                    4.84                4.76
1                        4.88                    4.82                4.71
2                        4.95                    4.85                4.64
3                        4.97                    4.89                4.83
4                        4.75                    5.00                5.00

  license instant_bookable calculated_host_listings_count  \
0     NaN              f                               2
1     NaN              f                               2
2     NaN              f                               1
3     NaN              f                               2
4     NaN              f                               2

  calculated_host_listings_count_entire_homes  \
0                                           0
1                                           1
2                                           1
3                                           0
4                                           2

  calculated_host_listings_count_private_rooms  \
0                                             2
1                                             1
2                                             0
3                                             2
4                                             0

  calculated_host_listings_count_shared_rooms reviews_per_month
0                                           0              2.63
1                                           0              0.39
2                                           0              0.65
3                                           0              2.68
4                                           0              0.02

[5 rows x 79 columns]
```

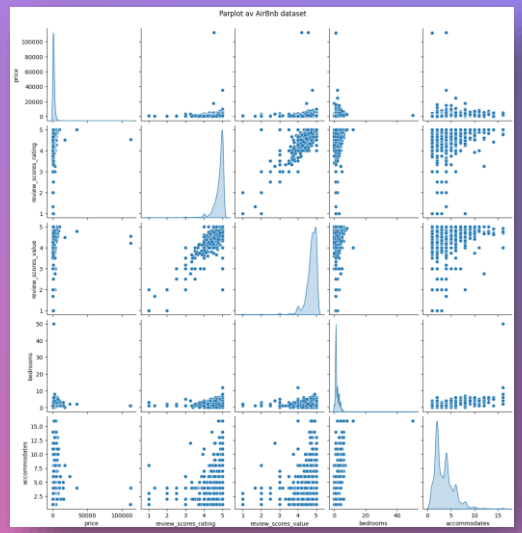
```
Statistik på alla numeriska kolumneriska kolumner (df.describe):
                 id     scrape_id       host_id host_listings_count  \
count  4.955000e+03  4.955000e+03  4.955000e+03          4951.000000
mean   6.984013e+17  2.025093e+13  1.978429e+08             7.068875
std    5.666998e+17  0.000000e+00  2.071420e+08            30.415664
min    1.644480e+05  2.025093e+13  4.457000e+03             1.000000
25%    3.730080e+07  2.025093e+13  2.449362e+07             1.000000
50%    8.745692e+17  2.025093e+13  1.003326e+08             1.000000
75%    1.181727e+18  2.025093e+13  3.595140e+08             2.000000
max    1.519717e+18  2.025093e+13  7.205346e+08          1456.000000

       host_total_listings_count  neighbourhood_group_cleansed     latitude  \
count                 4951.000000                           0.0  4955.000000
mean                   12.564532                           NaN    59.318008
std                   133.107379                           NaN     0.032222
min                     1.000000                           NaN    59.231990
25%                     1.000000                           NaN    59.297375
50%                     2.000000                           NaN    59.317130
75%                     4.000000                           NaN    59.337973
max                  8775.000000                           NaN    59.418950

          longitude  accommodates    bathrooms  ...  review_scores_checkin  \
count  4955.000000   4955.000000  3189.000000  ...            4085.000000
mean     18.030057      3.502523     1.287551  ...               4.860891
std       0.065606      2.023216     1.091029  ...               0.297532
min      17.773110      1.000000     0.000000  ...               1.000000
25%      17.999478      2.000000     1.000000  ...               4.840000
50%      18.046900      3.000000     1.000000  ...               4.950000
75%      18.076503      4.000000     1.500000  ...               5.000000
max      18.191930     16.000000    50.000000  ...               5.000000

       review_scores_communication  review_scores_location  \
count                  4085.000000             4085.000000
mean                      4.869435                4.810487
std                       0.308329                0.311399
min                       1.000000                1.000000
25%                       4.860000                4.750000
50%                       4.980000                4.980000
75%                       5.000000                5.000000
max                       5.000000                5.000000

       review_scores_value  license  calculated_host_listings_count  \
count          4085.000000      0.0                     4955.000000
mean              4.719907      NaN                        4.620182
std               0.366157      NaN                        9.952786
min               1.000000      NaN                        1.000000
25%               4.640000      NaN                        1.000000
50%               4.800000      NaN                        1.000000
75%               4.950000      NaN                        2.000000
max               5.000000      NaN                       51.000000

       calculated_host_listings_count_entire_homes  \
count                                  4955.000000
mean                                      4.040565
std                                       9.927759
min                                       0.000000
25%                                       1.000000
50%                                       1.000000
75%                                       1.000000
max                                      51.000000

       calculated_host_listings_count_private_rooms  \
count                                   4955.000000
mean                                       0.530575
std                                        1.945364
min                                        0.000000
25%                                        0.000000
50%                                        0.000000
75%                                        0.000000
max                                       21.000000

       calculated_host_listings_count_shared_rooms  reviews_per_month
count                                  4955.000000        4086.000000
mean                                      0.042987           1.280849
std                                       0.428554           2.162293
min                                       0.000000           0.010000
25%                                       0.000000           0.180000
50%                                       0.000000           0.500000
75%                                       0.000000           1.540000
max                                       6.000000          52.340000
```

```
Hur många observationer per område (df['neighbourhood'])
neighbourhood_cleansed
Södermalms                    1272
Normalm                        497
Hägersten-Liljeholmens         487
Kungsholmens                   470
Enskede-Årsta-Vantörs          456
Östermalms                     327
Bromma                         311
Skarpnäcks                     270
Älvsjö                         196
Farsta                         174
Hässelby-Vällingby             158
Rinkeby-Tensta                 135
Skärholmens                    105
Spånga-Tensta                   97
Name: count, dtype: int64
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 4955 entries, 0 to 4954
Data columns (total 79 columns):
 #   Column                                        Non-Null Count  Dtype
---  ------                                        --------------  -----
 0   id                                            4955 non-null   int64
 1   listing_url                                   4955 non-null   str
 2   scrape_id                                     4955 non-null   int64
 3   last_scraped                                  4955 non-null   str
 4   source                                        4955 non-null   str
 5   name                                          4955 non-null   str
 6   description                                   4763 non-null   str
 7   neighborhood_overview                         1780 non-null   str
 8   picture_url                                   4955 non-null   str
 9   host_id                                       4955 non-null   int64
 10  host_url                                      4955 non-null   str
 11  host_name                                     4951 non-null   str
 12  host_since                                    4951 non-null   str
 13  host_location                                 4815 non-null   str
 14  host_about                                    2242 non-null   str
 15  host_response_time                            3379 non-null   str
 16  host_response_rate                            3379 non-null   str
 17  host_acceptance_rate                          3801 non-null   str
 18  host_is_superhost                             4927 non-null   str
 19  host_thumbnail_url                            4951 non-null   str
 20  host_picture_url                              4951 non-null   str
 21  host_neighbourhood                            1370 non-null   str
 22  host_listings_count                           4951 non-null   float64
 23  host_total_listings_count                     4951 non-null   float64
 24  host_verifications                            4951 non-null   str
 25  host_has_profile_pic                          4951 non-null   str
 26  host_identity_verified                        4951 non-null   str
 27  neighbourhood                                 1780 non-null   str
 28  neighbourhood_cleansed                        4955 non-null   str
 29  neighbourhood_group_cleansed                  0 non-null      float64
 30  latitude                                      4955 non-null   float64
 31  longitude                                     4955 non-null   float64
 32  property_type                                 4955 non-null   str
 33  room_type                                     4955 non-null   str
 34  accommodates                                  4955 non-null   int64
 35  bathrooms                                     3189 non-null   float64
 36  bathrooms_text                                4947 non-null   str
 37  bedrooms                                      4697 non-null   float64
 38  beds                                          3188 non-null   float64
 39  amenities                                     4955 non-null   str
 40  price                                         3190 non-null   str
 41  minimum_nights                                4955 non-null   int64
 42  maximum_nights                                4955 non-null   int64
 43  minimum_minimum_nights                        4952 non-null   float64
 44  maximum_minimum_nights                        4952 non-null   float64
 45  minimum_maximum_nights                        4952 non-null   float64
 46  maximum_maximum_nights                        4952 non-null   float64
 47  minimum_nights_avg_ntm                        4955 non-null   float64
 48  maximum_nights_avg_ntm                        4955 non-null   float64
 49  calendar_updated                              0 non-null      float64
 50  has_availability                              4771 non-null   str
 51  availability_30                               4955 non-null   int64
 52  availability_60                               4955 non-null   int64
 53  availability_90                               4955 non-null   int64
 54  availability_365                              4955 non-null   int64
 55  calendar_last_scraped                         4955 non-null   str
 56  number_of_reviews                             4955 non-null   int64
 57  number_of_reviews_ltm                         4955 non-null   int64
 58  number_of_reviews_l30d                        4955 non-null   int64
 59  availability_eoy                              4955 non-null   int64
 60  number_of_reviews_ly                          4955 non-null   int64
 61  estimated_occupancy_l365d                     4955 non-null   int64
 62  estimated_revenue_l365d                       3190 non-null   float64
 63  first_review                                  4086 non-null   str
 64  last_review                                   4086 non-null   str
 65  review_scores_rating                          4086 non-null   float64
 66  review_scores_accuracy                        4086 non-null   float64
 67  review_scores_cleanliness                     4086 non-null   float64
 68  review_scores_checkin                         4085 non-null   float64
 69  review_scores_communication                   4085 non-null   float64
 70  review_scores_location                        4085 non-null   float64
 71  review_scores_value                           4085 non-null   float64
 72  license                                       0 non-null      float64
 73  instant_bookable                              4955 non-null   str
 74  calculated_host_listings_count                4955 non-null   int64
 75  calculated_host_listings_count_entire_homes   4955 non-null   int64
 76  calculated_host_listings_count_private_rooms  4955 non-null   int64
 77  calculated_host_listings_count_shared_rooms   4955 non-null   int64
 78  reviews_per_month                             4086 non-null   float64
dtypes: float64(20), int64(20), str(34)
memory usage: 3.0 MB
None
```

Datarening och EDA

Observationer och åtgärder – format och outliers

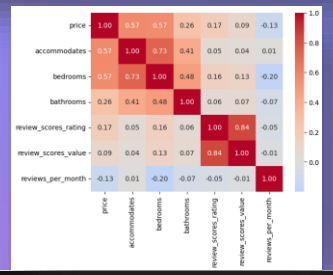
- Price är en sträng med formatet $123,45.00 – måste konverteras!
- Vi har några extrema outliers på pris, beror på datainmatningsfel. Beslutat att filtrera bort dessa poster med percentilcapping.
- Ett hotel hade registrerat hela sin kapacitet för rum och badrum. Droppade poster med >20 sovrum.
- Vi letar efter tomma poster med df.isna().sum() df.isna().sum()
- Gör Shapiro-testet för att se om pris är normalfördelat. Vid ej normalfördelning, gör priset logaritsmiskt vilket ger normalfördelade residualer som väsentligt förbättrar modellens prestanda.




```
0     $917.00
1     $450.00
2   $1,073.00
3     $779.00
4   $2,352.00
5   $1,126.00
6     $950.00
7     $751.00
8   $4,300.00
9   $1,218.00
Name: price, dtype: str
```

```python
# Städa och ändra price från str till int

cleandf = df
cleandf['price'] = cleandf['price'].str.replace('$','')
cleandf['price'] = cleandf['price'].str.replace(',','')
cleandf['price'] = cleandf['price'].astype(float)
```

```
count      3190.000000
mean       1690.473041
std        3704.741906
min         120.000000
25%         753.000000
50%        1200.000000
75%        2000.000000
max      112218.000000
Name: price, dtype: float64
```



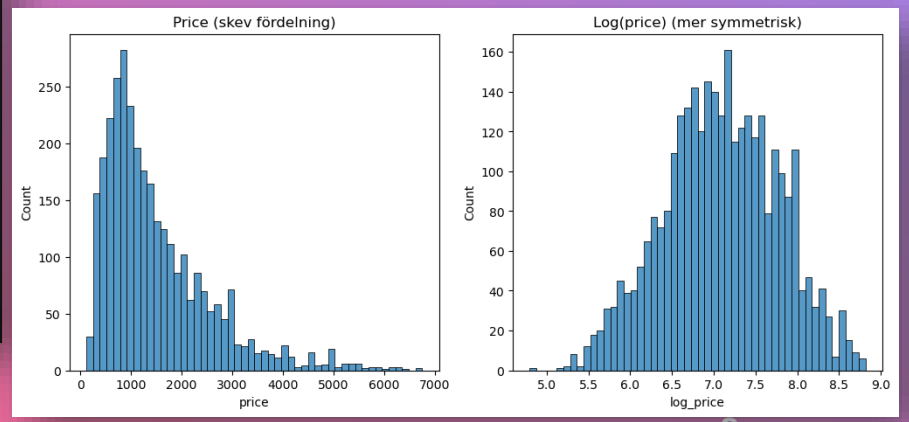

```python
# Percentilcapping. Vi behåller enbart de upp till 99 percentilen
upper = df["price"].quantile(0.99)

df = df[df["price"] <= upper].copy()
```

```python
df.isna().sum()
```

```
name                        0
listing_url                 0
price                       0
accommodates                0
bedrooms                    0
beds                        0
bathrooms                   1
property_type               0
room_type                   0
neighbourhood_cleansed      0
review_scores_rating      499
number_of_reviews           0
availability_365            0
minimum_nights              0
dtype: int64
```

```python
# Gör shapiro för att se om våra grupper av intresse är normalfördelade

from scipy.stats import shapiro

stat1, p1 = shapiro(cleandf['price'])

alpha = 0.05

if p1 < alpha:
    print(f"Grupp: Price. stat1: {stat1}. p-värde: {p1}. Vi kan förkasta nollhypotesen, inte normalfördelad")
else:
    print(f"Grupp: Price. stat1: {stat1}. p-värde: {p1}. Vi kan inte förkasta nollhypotesten, verkar normalfördelad"


# Gör pris logalgoritmiskt eftersom det inte är normalfördelat.
import numpy as np

cleandf['log_price'] = np.log1p(cleandf['price'])
```

```
Grupp: Price. stat1: 0.8614041344097955. p-värde: 3.4866202535568967e-46. Vi kan förkasta nollhypotesen, inte normal
fördelad
```



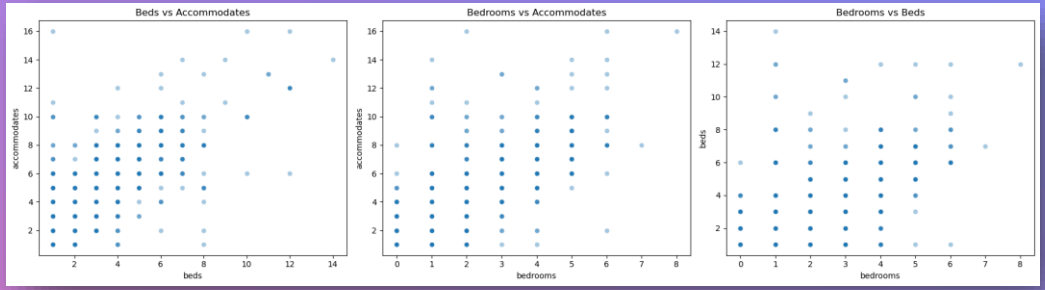
Datarening och EDA

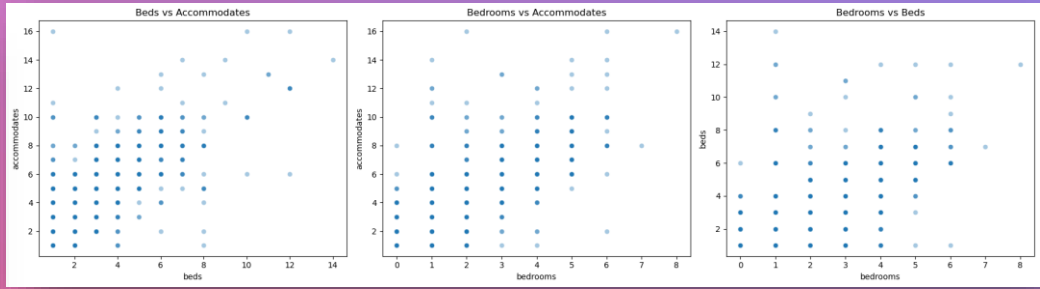
Observationer och åtgärder - boendestorlek

- Inkonsistent storleksdata. beds, bedrooms och accommodates hängde inte ihop logiskt. Exempel: 1 sovrum och 12 gäster, eller 0 sängar i ett boende. Inga direkt tillgängliga kvadratmeteruppgifter i datasetet.

- Orimliga relationer justerades med logiska gränser: Max 4 gäster per säng. Max 6 gäster per sovrum. Max 6 sängar per sovrum. Imputation.

- accommodates bedömdes mest tillförlitligt som storleksmått






```python
import numpy as np

# ersätt 0 för att undvika konstig division / multiplikation
beds_safe = df["beds"].replace(0, 1)
bedrooms_safe = df["bedrooms"].replace(0, 1)

# 1. max 4 gäster per säng
df["accommodates"] = np.where(
    df["accommodates"] > beds_safe * 4,
    beds_safe * 4,
    df["accommodates"]
)

# 2. max 6 gäster per sovrum
df["accommodates"] = np.where(
    df["accommodates"] > bedrooms_safe * 6,
    bedrooms_safe * 6,
    df["accommodates"]
)

# 3. max 6 sängar per sovrum
df["beds"] = np.where(
    df["beds"] > bedrooms_safe * 6,
    bedrooms_safe * 6,
    df["beds"]
)
```

Datarening och EDA

Observationer och åtgärder – shared rooms

- Shared rooms var missvisande.

- Manuell granskning av listningar visade: shared rooms = sängar i sovsalar på vandrarhem
- accommodates beskrev hela sovsalen (t.ex. 14 platser) istället för en bädd
- Ledde till skeva storleks- och prisrelationer.

- Varje rad representerar nu en faktisk bädd, inte hela sovsalen.

```
mask = df["room_type"] == "Shared room"

df.loc[mask, "accommodates"] = 1
df.loc[mask, "beds"] = 1
df.loc[mask, "bedrooms"] = 1
```

Datarening och EDA

Observationer och åtgärder – property group

- property_type-kolumnen hade för många varianter för att vara användbar direkt. Blandade boendetyp och uthyrningsform på ett svåranalyserat sätt.
- Lösningen: förenklad kategorisering med feature engineering.
- Resultat: Fyra tydliga grupper istället för ofantligt många varianter

```
property_type
Entire rental unit                    1720
Private room in rental unit            368
Entire home                            324
Entire condo                           186
Entire villa                            90
Room in hotel                           51
Private room in home                    49
Private room in condo                   47
Entire townhouse                        47
Entire serviced apartment               28
Private room in hostel                  26
Private room in bed and breakfast       26
Entire loft                             24
Room in aparthotel                      21
Tiny home                               21
Entire guesthouse                       20
Private room in villa                   12
Shared room in hotel                    10
Room in boutique hotel                  10
Private room in townhouse               10
Entire cabin                             9
Entire guest suite                       8
Shared room in hostel                    7
Shared room in rental unit               7
Camper/RV                                6
Private room in guest suite              4
Entire cottage                           4
Boat                                     3
Private room in guesthouse               3
Shared room in condo                     2
Private room in loft                     2
Casa particular                          1
Private room in casa particular          1
Private room in tiny home                1
Private room in serviced apartment       1
Shared room in bed and breakfast         1
Shared room in home                      1
Private room                             1
Private room in cabin                    1
Name: count, dtype: int64
```

```python
def simplify_property_type(x):
    x = x.lower()

    # 1. unique
    if any(word in x for word in ["boat", "camper"]):
        return "unique"

    # 2. guesthouse = house
    elif "guesthouse" in x:
        return "house"

    # 3. guest suite = apartment
    elif "guest suite" in x:
        return "apartment"

    # 4. vanliga house-typer
    elif any(word in x for word in [
        "home",
        "villa",
        "townhouse",
        "cabin",
        "cottage",
        "casa particular"
    ]):
        return "house"

    # 5. resten = apartment
    else:
        return "apartment"


df["property_group"] = df["property_type"].apply(simplify_property_type)

# shared rooms som egen kategori
df.loc[df["room_type"] == "Shared room", "property_group"] = "hostel"
```

```
property_group
apartment    2523
house         593
hostel         28
unique          9
Name: count, dtype: int64
```

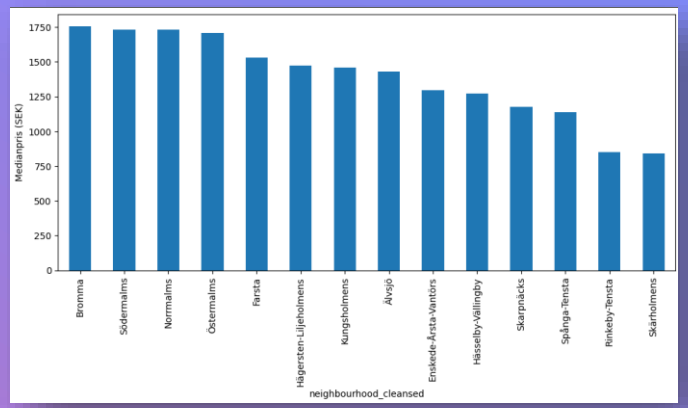
Datarening och EDA

Observationer och åtgärder – dyrt i förorten?

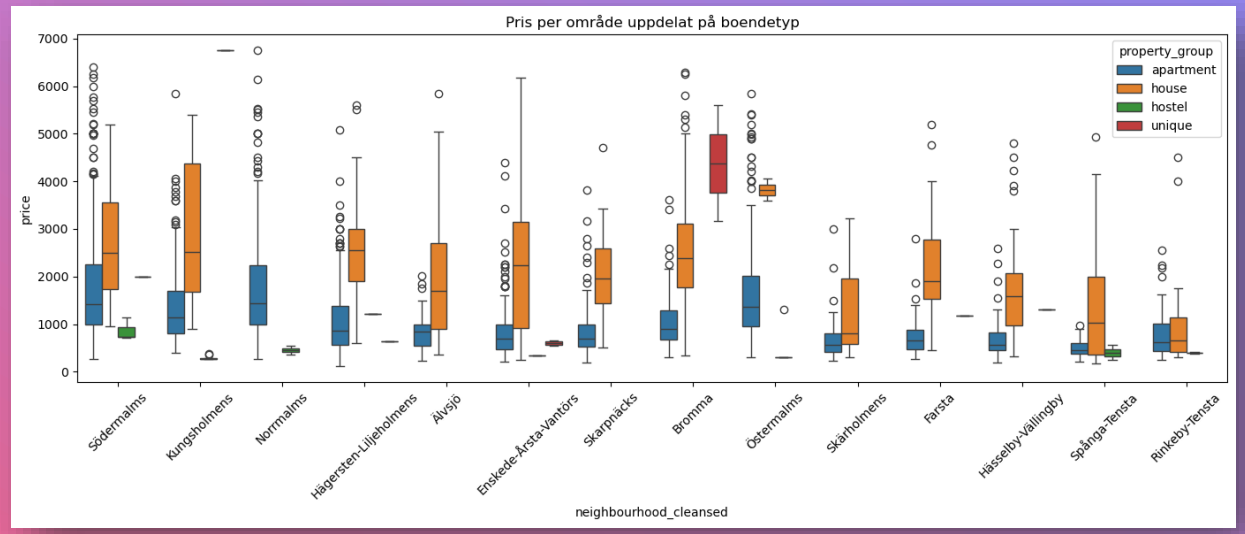
Varför verkade Farsta och Bromma lika dyra som innerstaden?

Boendetypen drog upp medelvärdet.
Farsta innehåller stora villor med pool nära sjöar.
När vi bryter ut boendetyp ser vi att lägenheter i Farsta ligger lågt.
Det är hustypen och inte läget som driver priset uppåt

Prisskillnader mellan områden förklaras till stor del av vad som hyrs ut, inte bara var.




```python
plt.figure(figsize=(14, 6))

sns.boxplot(
    data=df,
    x="neighbourhood_cleansed",
    y="price",
    hue="property_group"
)

plt.xticks(rotation=45)
plt.title("Pris per område uppdelat på boendetyp")
plt.tight_layout()
plt.show()
```



Datarening och EDA

One-Hot encoding & Regressionsmodell

Vi konverterar kategoriska variabler av intresse till numeriska med get_dummies så dessa kan hanteras av modellen, och drop_first=True är ofantligt viktigt för att undvika dummy variabel-fälla

Vi byggde fyra OLS-modeller och lade till en variabel i taget:
OLS 1 pris - accommodates
OLS 2+ neighbourhood
OLS 3+ room_type
OLS 4+ property_group

Varje tillägg förbättrade modellen
F-statistika: >150, p ≈ 0 – modellen som helhet är statistiskt signifikant

Nästan alla koefficienter har p < 0.05

```
cleandf = pd.get_dummies(
    cleandf,
    columns=["room_type", "neighbourhood_cleansed", "property_group"],
    drop_first=True
)
```

```
OLS1: Pris ~ accommodates
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.372
Model:                            OLS   Adj. R-squared:                  0.372
Method:                 Least Squares   F-statistic:                     1866.
Date:                Thu, 16 Apr 2026   Prob (F-statistic):          1.28e-320
Time:                        01:38:30   Log-Likelihood:                -2555.3
No. Observations:                3153   AIC:                             5115.
Df Residuals:                    3151   BIC:                             5127.
Df Model:                           1
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const            6.3586      0.019    326.855      0.000       6.321       6.397
accommodates     0.2126      0.005     43.199      0.000       0.203       0.222
==============================================================================
Omnibus:                        4.089   Durbin-Watson:                   1.840
Prob(Omnibus):                  0.129   Jarque-Bera (JB):                4.027
Skew:                           0.085   Prob(JB):                        0.134
Kurtosis:                       3.042   Cond. No.                         8.32
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

```
OLS2: + neighbourhood
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.501
Model:                            OLS   Adj. R-squared:                  0.499
Method:                 Least Squares   F-statistic:                     225.0
Date:                Thu, 16 Apr 2026   Prob (F-statistic):               0.00
Time:                        01:38:30   Log-Likelihood:                -2193.0
No. Observations:                3153   AIC:                             4416.
Df Residuals:                    3138   BIC:                             4507.
Df Model:                          14
Covariance Type:            nonrobust
==========================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                      6.2664      0.039    162.407      0.000       6.191       6.342
accommodates                               0.2208      0.005     49.016      0.000       0.212       0.230
neighbourhood_cleansed_Enskede-Årsta-Vantörs  -0.1744      0.044     -3.931      0.000      -0.261      -0.087
neighbourhood_cleansed_Farsta              -0.1038      0.057     -1.828      0.068      -0.215       0.008
neighbourhood_cleansed_Hägersten-Liljeholmens  -0.0554      0.044     -1.257      0.209      -0.142       0.031
neighbourhood_cleansed_Hässelby-Vällingby  -0.2814      0.057     -4.907      0.000      -0.394      -0.169
neighbourhood_cleansed_Kungsholmens        0.1083      0.044      2.470      0.014       0.022       0.194
neighbourhood_cleansed_Norrmalms           0.3094      0.044      7.090      0.000       0.224       0.395
neighbourhood_cleansed_Rinkeby-Tensta      -0.3089      0.058     -5.320      0.000      -0.423      -0.195
neighbourhood_cleansed_Skarpnäcks          -0.2357      0.051     -4.653      0.000      -0.335      -0.136
neighbourhood_cleansed_Skärholmens         -0.4108      0.063     -6.509      0.000      -0.535      -0.287
neighbourhood_cleansed_Spånga-Tensta       -0.3515      0.065     -5.446      0.000      -0.478      -0.225
neighbourhood_cleansed_Södermalms          0.3327      0.038      8.830      0.000       0.259       0.407
neighbourhood_cleansed_Älvsjö              -0.1296      0.055     -2.359      0.018      -0.237      -0.022
neighbourhood_cleansed_Östermalms          0.3146      0.048      6.603      0.000       0.221       0.408
==============================================================================
Omnibus:                       35.852   Durbin-Watson:                   1.849
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               41.167
Skew:                           0.204   Prob(JB):                     1.15e-09
Kurtosis:                       3.383   Cond. No.                         61.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

R² ökning: 0.372 → 0.501
```

```
OLS3: + room_type
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.535
Model:                            OLS   Adj. R-squared:                  0.532
Method:                 Least Squares   F-statistic:                     225.3
Date:                Thu, 16 Apr 2026   Prob (F-statistic):               0.00
Time:                        01:38:30   Log-Likelihood:                -2082.1
No. Observations:                3153   AIC:                             4198.
Df Residuals:                    3136   BIC:                             4301.
Df Model:                          16
Covariance Type:            nonrobust
==========================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                      6.4329      0.039    165.068      0.000       6.356       6.509
accommodates                               0.1911      0.005     39.895      0.000       0.182       0.201
neighbourhood_cleansed_Enskede-Årsta-Vantörs  -0.1428      0.043     -3.327      0.001      -0.227      -0.059
neighbourhood_cleansed_Farsta              -0.0872      0.055     -1.590      0.112      -0.195       0.020
neighbourhood_cleansed_Hägersten-Liljeholmens  -0.0600      0.043     -1.409      0.159      -0.143       0.024
neighbourhood_cleansed_Hässelby-Vällingby  -0.2542      0.055     -4.587      0.000      -0.363      -0.146
neighbourhood_cleansed_Kungsholmens        0.1245      0.042      2.938      0.003       0.041       0.208
neighbourhood_cleansed_Norrmalms           0.3142      0.042      7.452      0.000       0.232       0.397
neighbourhood_cleansed_Rinkeby-Tensta      -0.2649      0.056     -4.716      0.000      -0.375      -0.155
neighbourhood_cleansed_Skarpnäcks          -0.2188      0.049     -4.471      0.000      -0.315      -0.123
neighbourhood_cleansed_Skärholmens         -0.3638      0.061     -5.957      0.000      -0.483      -0.244
neighbourhood_cleansed_Spånga-Tensta       -0.2873      0.062     -4.598      0.000      -0.410      -0.165
neighbourhood_cleansed_Södermalms          0.3062      0.036      8.405      0.000       0.235       0.378
neighbourhood_cleansed_Älvsjö              -0.1142      0.053     -2.151      0.032      -0.218      -0.010
neighbourhood_cleansed_Östermalms          0.3030      0.046      6.578      0.000       0.213       0.393
room_type_Private room                     -0.3221      0.024    -13.660      0.000      -0.368      -0.276
room_type_Shared room                      -0.7121      0.091     -7.854      0.000      -0.890      -0.534
==============================================================================
Omnibus:                       66.292   Durbin-Watson:                   1.819
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               77.871
Skew:                           0.299   Prob(JB):                     1.23e-17
Kurtosis:                       3.485   Cond. No.                         61.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

R² ökning: 0.501 → 0.535
```

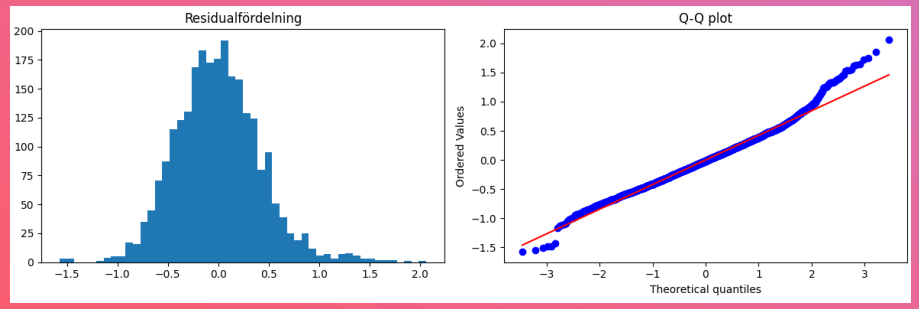
```
OLS4: + property_group
                            OLS Regression Results
==============================================================================
Dep. Variable:              log_price   R-squared:                       0.551
Model:                            OLS   Adj. R-squared:                  0.548
Method:                 Least Squares   F-statistic:                     213.6
Date:                Thu, 16 Apr 2026   Prob (F-statistic):               0.00
Time:                        06:53:07   Log-Likelihood:                -2026.4
No. Observations:                3153   AIC:                             4091.
Df Residuals:                    3134   BIC:                             4206.
Df Model:                          18
Covariance Type:            nonrobust
==========================================================================================
                                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                      6.4217      0.038    167.671      0.000       6.347       6.497
accommodates                               0.1597      0.005     30.916      0.000       0.150       0.170
neighbourhood_cleansed_Enskede-Årsta-Vantörs  -0.1234      0.042     -2.922      0.004      -0.206      -0.041
neighbourhood_cleansed_Farsta              -0.1312      0.054     -2.428      0.015      -0.237      -0.025
neighbourhood_cleansed_Hägersten-Liljeholmens  -0.0124      0.042     -0.296      0.767      -0.095       0.070
neighbourhood_cleansed_Hässelby-Vällingby  -0.2925      0.055     -5.362      0.000      -0.399      -0.186
neighbourhood_cleansed_Kungsholmens        0.2216      0.042      5.215      0.000       0.138       0.305
neighbourhood_cleansed_Norrmalms           0.4360      0.043     10.252      0.000       0.353       0.519
neighbourhood_cleansed_Rinkeby-Tensta      -0.2389      0.055     -4.323      0.000      -0.347      -0.131
neighbourhood_cleansed_Skarpnäcks          -0.1001      0.048     -3.735      0.000      -0.275      -0.086
neighbourhood_cleansed_Skärholmens         -0.3497      0.060     -5.825      0.000      -0.467      -0.232
neighbourhood_cleansed_Spånga-Tensta       -0.3943      0.062     -6.355      0.000      -0.516      -0.273
neighbourhood_cleansed_Södermalms          0.4195      0.037     11.336      0.000       0.347       0.492
neighbourhood_cleansed_Älvsjö              -0.1635      0.052     -3.123      0.002      -0.266      -0.061
neighbourhood_cleansed_Östermalms          0.4005      0.046      8.682      0.000       0.310       0.491
room_type_Private room                     -0.3427      0.023    -14.794      0.000      -0.388      -0.297
room_type_Shared room                      -0.5442      0.044    -12.322      0.000      -0.631      -0.458
property_group_hostel                      -0.5442      0.044    -12.322      0.000      -0.631      -0.458
property_group_house                       0.3449      0.027     12.563      0.000       0.291       0.399
property_group_unique                      0.2016      0.155      1.302      0.193      -0.102       0.505
==============================================================================
Omnibus:                       68.618   Durbin-Watson:                   1.818
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               79.439
Skew:                           0.311   Prob(JB):                     5.62e-18
Kurtosis:                       3.466   Cond. No.                     2.84e+16
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 6.51e-29. This might indicate that there are
strong multicollinearity problems or that the design matrix is singular.
```

Datarening och EDA

Train/Test

Upplägg: 80% träning, 20% test (osedd data)

TolkningR^2 träning ≈ R^2 test → ingen overfitting
Modellen generaliserar bra på data den aldrig sett
I snitt gissar modellen fel med ~500 kr per natt




```python
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score
import numpy as np

# ===== 1. Kopiera din rensade data =====
cleandf = df.copy()

# ===== 2. One-hot encode (Emil + din property_group) =====
cleandf = pd.get_dummies(
    cleandf,
    columns=["room_type", "neighbourhood_cleansed", "property_group"],
    drop_first=True
)

# ===== 3. Hitta dummy-kolumner =====
neighbourhood_dummies = [col for col in cleandf.columns if col.startswith("neighbourhood_cleansed_")]
room_dummies = [col for col in cleandf.columns if col.startswith("room_type_")]
property_dummies = [col for col in cleandf.columns if col.startswith("property_group_")]

# ===== 4. Features =====
features = [
    "accommodates",
    "bedrooms",
    "beds",
    "bathrooms",
    "review_scores_rating",
    "number_of_reviews",
    "availability_365",
    "minimum_nights",
    "has_rating"
] + neighbourhood_dummies + room_dummies + property_dummies

# ===== 5. Target (log) =====
cleandf["log_price"] = np.log1p(cleandf["price"])

X = cleandf[features].astype(float)
y = cleandf["log_price"]

# ===== 6. Train/test =====
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

X_train_sm = sm.add_constant(X_train)
X_test_sm = sm.add_constant(X_test)

# ===== 7. Modell =====
model_you = sm.OLS(y_train, X_train_sm).fit()

# ===== 8. Prediktion =====
y_pred = model_you.predict(X_test_sm)

# ===== 9. Utvärdering =====
r2_train = r2_score(y_train, model_you.predict(X_train_sm))
r2_test = r2_score(y_test, y_pred)

mae_log = mean_absolute_error(y_test, y_pred)
mae_sek = mean_absolute_error(np.expm1(y_test), np.expm1(y_pred))

print("=== MIN MODELL ===")
print(f"R² träning: {r2_train:.3f}")
print(f"R² test:    {r2_test:.3f}")
print(f"MAE (log):  {mae_log:.3f}")
print(f"MAE (SEK):  {mae_sek:.0f} kr")
```

```
=== MIN MODELL ===
R² träning: 0.620
R² test:    0.649
MAE (log):  0.323
MAE (SEK):  496 kr
```

Datarening och EDA

Extra

```
TOP 10 ÖVERPRISADE:
                                                name  neighbourhood_cleansed
4216                             Room with a nice view   Enskede-Årsta-Vantörs
1584                  Riddersviks Herrgård, Svit Delux      Hässelby-Vällingby
1410                        Healing Camp - Mind & Body      Hässelby-Vällingby
1220                          Generator - Studio Suite               Norrmalms
1438                        Riddersviks Herrgård, Svit      Hässelby-Vällingby
4885                          Generator - Studio Suite               Norrmalms
2078                  Sekelskiftesfyra vid Mariatorget              Södermalms
1437                   Riddersviks Herrgård, Dubbelrum      Hässelby-Vällingby
902                     Top floor, Old Town, Stockholm              Södermalms
501   A cozy apartment 13 minutes from Stockholm City  Hägersten-Liljeholmens

         room_type  accommodates   price  predicted_price    pct_diff  \
4216  Private room           1.0  4400.0       552.231784  696.766888
1584  Private room           3.0  2588.0       424.008404  510.365260
1410  Private room           4.0  3913.0       677.041838  477.955420
1220  Private room           2.0  6138.0      1105.297297  455.325705
1438  Private room           4.0  2271.0       439.129314  417.159736
4885  Private room           2.0  5524.0      1104.587569  400.096158
2078  Private room           4.0  6400.0      1317.970199  385.595198
1437  Private room           3.0  1898.0       403.230943  370.698004
902   Private room           2.0  5000.0      1081.350232  362.384883
501   Private room           2.0  3000.0       652.488259  359.778388
```

```
TOP 10 BÄSTA DEALS - billigare än modellen förväntar sig:
                                                name  neighbourhood_cleansed
2271                                        tiny home  Hägersten-Liljeholmens
3203                  Extra övernattnings rum i stan               Norrmalms
3757                        Stureby(södra stockholm)   Enskede-Årsta-Vantörs
4851       Mysig etta på Mariatorget! Uthyres 1-31/9              Södermalms
2291                          mycket trevlig lägenhet   Enskede-Årsta-Vantörs
4951        Furnished Room in the Heart of Stockholm              Södermalms
4442                Rymlig villa nära stad och natur                  Älvsjö
4229            Rymligt hus med 5 sovrum i Stockholm            Kungsholmens
4774  Workers House • 6BR • 2Bath • 12 beds • All-inc      Hässelby-Vällingby
4157                     Rum i bostadsrätt i Vasastan              Östermalms

            room_type  accommodates   price  predicted_price   pct_diff  \
2271     Private room           1.0   120.0       557.616310 -78.479826
3203  Entire home/apt           1.0   270.0      1201.358872 -77.525450
3757  Entire home/apt           4.0   232.0       997.591360 -76.743985
4851  Entire home/apt           2.0   347.0      1463.085329 -76.282996
2291  Entire home/apt           2.0   372.0      1552.464528 -76.038100
4951     Private room           2.0   272.0      1128.257077 -75.892019
4442  Entire home/apt          16.0  2743.0     11286.595229 -75.696834
4229  Entire home/apt          10.0  1658.0      5851.600810 -71.665873
4774  Entire home/apt          12.0  1586.0      5001.821195 -68.291549
4157     Private room           1.0   299.0       941.202675 -68.232134

                                          listing_url
2271   https://www.airbnb.com/rooms/792433644479186644
3203  https://www.airbnb.com/rooms/1088167261137903056
3757  https://www.airbnb.com/rooms/1187777569265003996
4851  https://www.airbnb.com/rooms/1491986701179034489
```